Mathematical Foundations

CMPSC 133 - Design and Analysis of Algorithms

What are Algorithms?

- Procedure used for solving a problem or performing a calculation.
- Composed of exact list of instructions that conduct specified actions step-by-step in hardware or software-based routines.
- Example of algorithms in everyday life:
  - Following recipes (e.g., frying a sunny-side-up egg)
  - Tying shoelaces (e.g., bunny rabbit)
  - Sorting paper by sizes
  - Searching for books in libraries

(Gillis, 2024)

Example: The Bunny Rabbit Method

START

1. Take a shoelace in each hand.
2. Cross the right lace over the left lace halfway up the laces.
3. Take the end of the right lace and put it over the left lace.
4. Poke the right lace underneath the red lace.
5. Hold on to the ends of both laces and pull tight.
6. Make small loops for both laces, resembling "bunny ears."
7. Cross the middle of the right lace's bunny ear over the middle of the left lace's bunny ear.
8. Take the top of the left lace's bunny ear and put it over the right lace's bunny ear.
9. Poke the left lace's bunny ear through the hole created by Step 7.
10. Put your fingers inside the ends of the bunny ears, then hold on and pull tight

Example: The Bunny Rabbit Method

END

(The GEM Foundation, n.d.)

Algorithms in the Context of Computer Science

- Small procedure that solves a recurrent ~~problem.~~ adj. occurring often or repeatedly (Oxford Languages, n.d.).
- Used as specifications for performing data processing and play a major role in automated systems.
- Includes an initial input ~~along with a set of instructions.~~ initial data needed to make decisions.

  can be represented in the form of numbers or words.

(Gillis, 2024)

Different Types of Algorithms

- Search Engine Algorithm
  - Input: Search strings or keywords, operators.
  - Searches the associated database for relevant webpages and returns results.
- Encryption and/or Decryption Algorithm
  - Input: Data to encrypt / decrypt, (in some instances,) key.
  - Transforms data according to specified actions to protect its contents.
- Hashing Algorithm
  - Input: Data
  - Takes data and converts it to some uniform message with a hashing

(Gillis, 2024)

References

In order of appearance:

- Gillis, A. S. (2024, July 29). *What is An Algorithm?* What is an algorithm? https://www.techtarget.com/whatis/definition/algorithm
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- Oxford Dictionaries. (n.d.). Recurrent. Define Recurrent - Google Search. https://www.google.com/search?q=define%2Brecurrent&oq=define%2Brecurrent&gs_lcrp=EgRIZGdlKgYIABBFGEAyBggAEEUYQDINCAEQABiDARixAxiABDIHCAIQABiABDIHCAMQABiABDIHCAQQABiABDINCAUQABiDARixAxiABDIHCAYQABiABDIHCAcQABiABDIHCAgQABiABNIBCTEwNTU0ajBqMagCALACAA&sourceid=chrome&ie=UTF-8